

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

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1. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyse system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.

1. Introduction

An organization can develop a spam detection system using probabilistic machine learning. The system classifies an email as Spam or Not Spam (Ham) based on features such as words, links, sender information and attachments. Naive Bayes and Bayes Optimal Classifier are suitable because they can make predictions even when there is uncertainty in the data.

2. Naive Bayes Classifier

Naive Bayes applies Bayes Theorem to calculate the probability of an email belonging to each class.

Bayes Theorem: $P(\text{Class} \mid \text{Features}) = P(\text{Features} \mid \text{Class}) \times P(\text{Class}) / P(\text{Features})$

For spam detection: $P(\text{Spam} \mid \text{Features}) = P(\text{Features} \mid \text{Spam}) \times P(\text{Spam}) / P(\text{Features})$

The same probability is calculated for Ham. The class with the higher probability is selected.

Naive Bayes assumes that the features are conditionally independent given the class.

3. Features Used

The system can use:

- Spam-related words such as "free", "offer" and "winner"
- Number of links
- Number of attachments
- Sender information
- Email length

4. Bayes Optimal Classifier

Bayes Optimal Classifier considers different possible hypotheses and uses their posterior probabilities to make the final decision. It provides a theoretically optimal classification when the correct probability distribution is known.

It can be represented as: $P(\text{Class} \mid \mathbf{X}) = \text{Sum of } [P(\text{Class} \mid \mathbf{h}, \mathbf{X}) \times P(\mathbf{h} \mid \mathbf{X})]$

where:

- X = observed features
- h = possible hypothesis

This approach handles uncertainty better than selecting only one hypothesis.

5. MLE and MDL Comparison

Maximum Likelihood Estimation (MLE)	Minimum Description Length (MDL)
Finds parameters that best fit the training data	Selects a model with minimum total description length
Focuses mainly on likelihood	Considers data and model complexity
Can produce complex models	Prefers simpler models
May suffer from overfitting	Helps reduce overfitting
Useful for estimating probabilities	Useful for feature/model selection

6. Sample Size and Hypothesis Space

A larger training dataset provides better probability estimates and generally improves classification accuracy.

With a small sample:

- Probability estimates may be unreliable.
- Rare spam patterns may not be learned.
- Classification errors may increase.

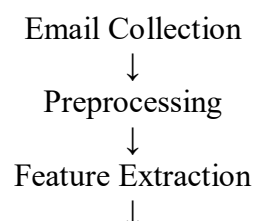
The hypothesis space contains possible classification rules or models. A very large hypothesis space can increase complexity and require more training data. A smaller hypothesis space is easier to learn but may not represent all spam patterns.

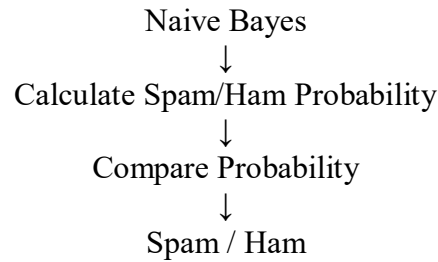
7. Performance Evaluation

The system can be evaluated using:

- Accuracy
- Precision
- Recall
- F1-score

8. Flowchart





9. Conclusion

Naive Bayes provides a simple and efficient spam classification method. Bayes Optimal Classifier improves decision-making by considering uncertainty among hypotheses. MLE focuses on fitting the data, while MDL balances accuracy and model complexity. A sufficiently large sample size and suitable hypothesis space are important for good performance.

2. An autonomous vehicle system must make real-time driving decisions under uncertain environmental conditions such as weather, traffic, and pedestrian movement. Using Bayesian inference, Naïve Bayes classifiers, and probabilistic reasoning, design a model to predict safe driving actions. Analyse how uncertainty and sensor noise are handled, and evaluate system reliability under dynamic conditions.

1. Introduction

An autonomous vehicle must make real-time decisions under uncertain environmental conditions. Weather, traffic, pedestrians, road conditions and sensor noise can affect its decisions. Bayesian inference, Naive Bayes and probabilistic reasoning can be used to select safe driving actions.

2. Input Data

The vehicle can collect information from:

- Camera
- Radar
- LiDAR
- GPS
- Weather sensors

3. Bayesian Inference

Bayesian inference calculates the probability of a situation based on available evidence.

Formula: $P(H | E) = P(E | H) \times P(H) / P(E)$

where:

- H = hypothesis
- E = evidence

4. Naive Bayes Classification

Naive Bayes can classify the current driving situation into different actions:

Situation	Action
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Clear road	Continue driving
Heavy traffic	Slow down
Pedestrian detected	Slow down or stop
Obstacle detected	Emergency braking
Poor visibility	Reduce speed

5. Handling Uncertainty and Sensor Noise

Sensor readings may be incorrect because of:

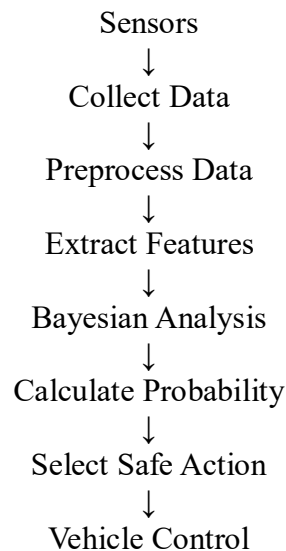
- Rain
- Fog
- Darkness
- Sensor failure
- Poor sensor position

The system can combine information from multiple sensors instead of depending on a single sensor. Probability values can also represent the confidence of each observation.

6. Probabilistic Decision Making

The system continuously receives new information and updates its probabilities. If traffic suddenly increases, the probability of an unsafe driving condition increases and the vehicle can reduce speed.

7. System Flowchart



8. Reliability

The reliability of the system depends on:

- Sensor accuracy
- Training data quality

- Correct probability estimation
- Fast computation
- Proper handling of sensor noise

9. Advantages

- Handles uncertain information.
- Combines multiple sensor observations.
- Supports real-time decisions.
- Can update predictions when new data arrives.
- Helps improve driving safety.

10. Conclusion

Bayesian inference and Naive Bayes provide useful probabilistic methods for autonomous vehicles. They allow the vehicle to handle uncertainty, combine sensor information and select safer actions in changing environments.

3. An e-commerce platform aims to recommend products to users based on browsing and purchase history. Using Naïve Bayes and Bayesian learning concepts, design a recommendation model. Analyse how conditional probabilities influence recommendations and how uncertainty in user preferences is managed. Evaluate scalability and performance in handling large, dynamic datasets.

1. Introduction

An e-commerce platform can recommend products by analysing user browsing and purchase history. Naive Bayes and Bayesian learning can estimate the probability that a user will be interested in a particular product.

2. Data Used

The system can use:

- Browsing history
- Purchase history
- Search history
- Products viewed
- Product ratings

3. Naive Bayes Model

The system calculates the probability of a product based on the user's features.

Formula: $P(\text{Product} \mid \text{Features}) = P(\text{Features} \mid \text{Product}) \times P(\text{Product}) / P(\text{Features})$

Products with higher probabilities are ranked higher and recommended to the user.

4. Conditional Probabilities

Conditional probability helps identify relationships between user actions and products.

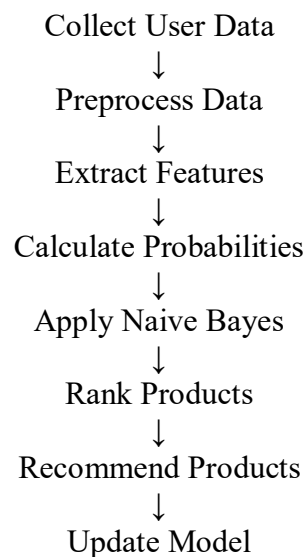
For example: **$P(\text{Purchase} \mid \text{Product Viewed})$**

If users who view mobile phones frequently purchase mobile accessories, the system can use this relationship to recommend accessories.

5. Handling Uncertainty

User preferences are uncertain because a user may browse a product without purchasing it or may change interests over time. Bayesian learning represents these preferences using probabilities instead of fixed rules. When new browsing or purchase data becomes available, the probabilities can be updated.

6. Recommendation Process



7. Scalability and Performance

Large e-commerce platforms may have millions of users and products. The system should use efficient algorithms and data processing techniques.

Performance can be measured using:

Metric	Purpose
Accuracy	Measures prediction correctness
Precision	Measures relevance of recommendations
Recall	Measures relevant products successfully recommended
Click-through Rate	Measures user interaction
Conversion Rate	Measures purchase from recommendations

8. Advantages

- Simple and fast.
- Handles uncertainty.
- Can work with many features.
- Easy to update.

- Suitable for large datasets with efficient implementation.

9. Conclusion

Naive Bayes provides a simple probabilistic approach for product recommendation. Conditional probabilities help understand user behaviour, while Bayesian learning allows recommendations to change as new user information becomes available.

4. A smart healthcare system is designed to monitor patients in real-time using wearable sensors. Using Bayesian Belief Networks and probability learning, develop a model to detect anomalies in vital signs. Analyse how conditional dependencies between physiological parameters are modelled and how uncertainty due to sensor noise is handled. Evaluate system performance and discuss challenges in real-time deployment.

1. Introduction

A smart healthcare system can use wearable sensors to monitor patients continuously. Sensors can collect heart rate, blood pressure, temperature and oxygen levels. Bayesian Belief Networks and probability learning can be used to identify abnormal health conditions.

2. Sensor Data

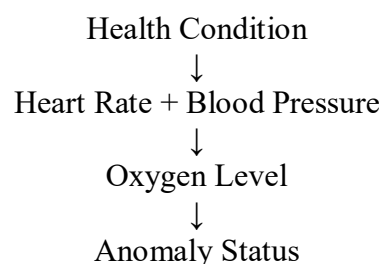
The system monitors:

- Heart rate
- Blood pressure
- Body temperature
- Oxygen saturation
- Respiratory rate

3. Bayesian Belief Network

A Bayesian Belief Network represents relationships between different variables.

Example:



For example, a health condition may affect heart rate and blood pressure. These parameters can provide evidence about the patient's condition.

4. Bayesian Probability

The system calculates: $P(\text{Anomaly} \mid \text{Evidence})$

Using Bayes Theorem: $P(\text{Anomaly} \mid \text{Evidence}) = P(\text{Evidence} \mid \text{Anomaly}) \times P(\text{Anomaly}) / P(\text{Evidence})$

If the calculated probability is high, the system can generate an alert.

5. Conditional Dependencies

Physiological parameters are not completely independent.

For example:

- A health condition may increase heart rate.
- A health condition may affect blood pressure.
- Abnormal oxygen levels may provide additional evidence.
- Multiple abnormal readings can increase the probability of an anomaly.

6. Handling Sensor Noise

Wearable sensors can produce incorrect readings because of:

- Patient movement
- Poor sensor contact
- Sensor malfunction
- Network problems

7. Probability Learning

Historical patient data can be used to learn probability values.

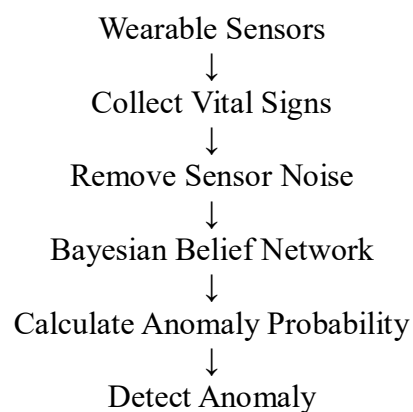
For example: **$P(\text{High Heart Rate} \mid \text{Anomaly})$**

The system can update these probabilities as more patient data becomes available.

8. Performance Evaluation

Measure	Purpose
Accuracy	Correct anomaly detection
Precision	Reduces false alarms
Recall	Detects actual abnormalities
Response Time	Measures speed of alert generation

9. Real-Time Flowchart



↓
Generate Alert

10. Challenges

- Continuous data processing
- Sensor failure
- False alarms
- Network connectivity
- Battery limitations
- Data privacy

11. Conclusion

Bayesian Belief Networks can effectively model relationships between physiological parameters and detect health anomalies. Probability learning improves the model using historical data, while sensor filtering helps handle noisy readings. The system can provide useful real-time healthcare alerts.

5. An agriculture-based system aims to predict crop yield under uncertain environmental conditions such as rainfall, soil quality, and temperature. Using Bayesian inference and probability learning, design a predictive model. Analyse how prior agricultural knowledge and seasonal variations influence predictions. Evaluate the model's reliability and discuss its impact on decision-making for farmers and policy planning.

1. Introduction

Crop yield depends on uncertain environmental conditions such as rainfall, soil quality and temperature. Bayesian inference and probability learning can be used to predict crop yield and support agricultural decision-making.

2. Input Factors

The model can use:

- Rainfall
- Temperature
- Soil quality
- Irrigation
- Fertilizer usage
- Crop type

3. Bayesian Inference

Bayesian inference combines previous agricultural knowledge with current environmental information.

Formula: $P(\text{Yield} \mid \text{Evidence}) = P(\text{Evidence} \mid \text{Yield}) \times P(\text{Yield}) / P(\text{Evidence})$

where:

- Yield = crop yield
- Evidence = rainfall, temperature, soil quality and other factors
- $P(\text{Yield})$ = prior probability
- $P(\text{Yield} | \text{Evidence})$ = posterior probability

4. Role of Prior Knowledge

Prior agricultural knowledge can come from:

- Historical crop yield
- Previous rainfall records
- Soil information
- Agricultural expert knowledge
- Seasonal information

5. Probability Learning

The system learns relationships between environmental conditions and crop yield.

Examples:

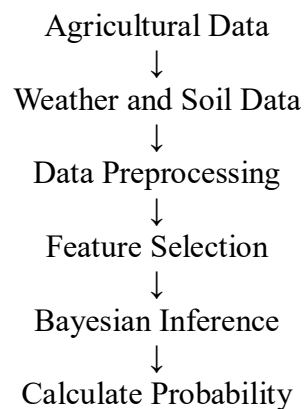
- $P(\text{High Yield} | \text{Good Soil})$
- $P(\text{High Yield} | \text{Adequate Rainfall})$
- $P(\text{Low Yield} | \text{Low Rainfall})$

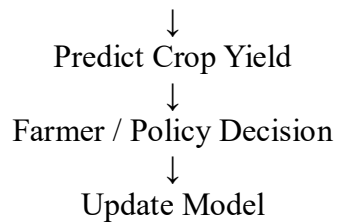
6. Seasonal Variations

Seasonal conditions affect crop growth.

Condition	Expected Effect
Good soil + adequate rainfall	High yield
Good soil + low rainfall	Medium yield
Poor soil + adequate rainfall	Medium yield
Poor soil + low rainfall	Low yield
High temperature + low rainfall	Low yield

7. Prediction Flowchart





8. Reliability Evaluation

The model can be evaluated by comparing predicted yield with actual yield.

Important measures include:

- Prediction accuracy
- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- Confidence of prediction

9. Impact on Farmers

The system can help farmers:

- Select suitable crops.
- Plan irrigation.
- Manage fertilizers.
- Estimate expected production.

10. Impact on Policy Planning

Government organizations can use crop yield predictions for:

- Food supply planning
- Irrigation planning
- Crop insurance
- Agricultural subsidies
- Food security planning

11. Conclusion

Bayesian inference provides a useful method for crop yield prediction under uncertain environmental conditions. By combining prior agricultural knowledge with current data and continuously updating probabilities, the system can support farmers and policymakers in making better agricultural decisions.